kata – Describe katas with pictograms

Lukas Heindl*

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Abstract

The `kata` package provides a compact and structured way to typeset martial arts katas[1] (forms) in LaTeX. It models a kata as a sequence of steps, each combining the stance hand/foot technique, and transition direction, and supports customizing the resulting pictograms for clear visual representation.

1 Introduction

In many martial arts such as Karate, a kata is a predefined sequence of movements[2].

The `kata` package aims to provide a simple, structured way to describe katas in LaTeX. Each step of a kata is expressed as a step consisting of

1. the stance with direction
2. a technique with direction
3. the direction of movement or rotation into the next step

The package also offers customizing the resulting pictograms, allowing visual annotation of sequences, such as highlighting repetitions or emphasizing specific actions.

2 Usage / Quick-Start

Load this package as `\usepackage[<options>]{kata}`.

This section documents every command and environment which mostly is all you need to know to start using this package. Together with the simple example (see section 2.3) this should get you working.

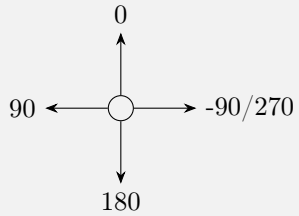
Then, you can look up customization options later in section 2.4 (either by reading everything or following the links embedded in this section).

*E-Mail: oss.heindl+latex@protonmail.com
Issue tracker: github.com/atticus-sullivan/kata/issues
Github (mirror): github.com/atticus-sullivan/kata

[1] https://en.wikipedia.org/wiki/Kata
[2] https://en.wikipedia.org/wiki/Kata

Note: Orientation of the *angles* in the arguments

Some commands such as `\kataWaza`[→P. 3], `\kataDir`[→P. 4], and `\kataStance`[→P. 4] can/must be passed angles in the range of -360 to 360 degrees. Also some pgfkeys take a angles as an argument.
This package consistently defines the orientation of these angles as follows:



2.1 Package arguments

This package supports passing package-options.
See section 2.4 for the available ⟨*options*⟩.

`\kataSetup`[⟨*options*⟩]

This can set exactly the same options like with the package-options. You can either use this to set package-options at a later point in time in the preamble. But you can also use this to set options mid-document. Note that these options are then local to that scope. You can leverage this to change the style of a sequence of `kataStep`[→P. 3]s "globally" (see also section 2.5).

`\kataAliases`

Defines the following aliases for ease of use:

`\stance`[⟨*side*⟩]{⟨*content*⟩}{⟨*dir*⟩}[⟨*options*⟩]

an alias for `\kataStance`[→P. 4]

`\waza`[⟨*side*⟩]{⟨*content*⟩}{⟨*dir*⟩}[⟨*options*⟩]

an alias for `\kataWaza`[→P. 3]

`\dir`[⟨*options*⟩]{⟨*angleA*⟩}{⟨*angleB*⟩}

an alias for `\kataDir`[→P. 4]

`\kiai`{⟨*angle*⟩}

an alias for `\kataKiai`[→P. 4]

```
\begin{step}[⟨options⟩]
  ⟨environment content⟩
\end{step}
```

an alias for `kataStep`[→P. 3]

Defining the aliases is deferred until the begin of the document. Each alias is only defined if the corresponding macro is not already defined (otherwise a warning is shown in the log)
At the moment, this option only has an effect when given in the preamble (via `\kataSetup`) or in the package-options.

2.2 Specifying a kata

```
\begin{kata}[⟨options⟩]{⟨title⟩}{⟨margin⟩}[⟨belt-color⟩]
  ⟨environment content⟩
\end{kata}
```

The environment which holds an entire kata. Within it, normal LaTeX code can be typeset. Mostly the content usually consists of `kataStep` environments each representing one step in the kata.
This will also add a section named ⟨*title*⟩.
It also adds a marginline which is intended to indicate for which belt this kata is required. Thus, the marginline consists of

- an arrow pointing down
- ⟨*margin*⟩
- a drawing showing a belt. The color by default will be ⟨*margin*⟩ but it may be overwritten with ⟨*belt-color*⟩ If ⟨*margin*⟩ is empty and ⟨*belt-color*⟩ is not given, no marginline will be drawn

See section 2.4 for the available ⟨*options*⟩.
Note: After all steps have been drawn, it will automatically add a number of *phantom*-`kataSteps` (see `/kata/setup/step phantom style` → P. 7 and `/kata/step/phantom` → P. 8) trying to generate a uniform looking grid of pictograms. For this to work correctly, `/kata/kata/width` → P. 8 needs be set to the number of pictograms fitting one line on the page.

```
\begin{kataStep}[⟨options⟩]
  ⟨environment content⟩
\end{kataStep}
```

Draws a pictogram representing one step in the kata.
See section 2.4 for the available ⟨*options*⟩.
The body will be the content of a `tikzpicture` so normal LaTeX code won't work as expected. Instead, if you want/need to, you can draw arbitraty TikZ code in this environment.
Note: The circle representing the *head*, is drawn right at the start of this environment as a node named *head*. You may use it as a regular TikZ node when writing custom TikZ-code. The advice is not to draw out of the normal bounds as this will break the alignment of the pictograms.
Usually you'll use `\kataWaza`, `\kataStance` → P. 4 and `\kataDir` → P. 4 to populate the step with contents. Note that all of these commands may be present (multiple times) but they don't need to be present all.

```
\kataWaza[⟨side⟩]{⟨content⟩}{⟨dir⟩}[⟨options⟩]
```

Use this command to draw a technique with its direction.
Note: This command is quite similar to `\kataStance` → P. 4.

⟨***side***⟩ may specify the side (R/L or right/left) which executes the technique (arbitrary text is allowed here)

⟨***content***⟩ the text describing the action

⟨***angle***⟩ the angle in which to execute the action. See section 2 for more details.

⟨***options***⟩ see section 2.4.5 for the available ⟨*options*⟩. The most common option probably here is `/kata/waza/flip` → P. 9.

`\kataStance[⟨side⟩]{⟨content⟩}{⟨dir⟩}[⟨options⟩]`

Use this command to draw the technique+direction for the stance.
Note: This command is quite similar to `\kataWaza` → P. 3.

⟨***side***⟩ may specify the side (R/L or right/left) which executes the technique (arbitrary text is allowed here)

⟨***angle***⟩ the angle in which to execute the action. See section 2 for more details.

⟨***dir***⟩ the direction in which to execute the action

⟨***options***⟩ see section 2.4.5 for the available ⟨*options*⟩. The most common option probably here is `/kata/stance/flip` → P. 9.

`\kataDir[⟨options⟩]{⟨angleA⟩}{⟨angleB⟩}`

Use this command to draw an indicator indicating the direction/turn to transition to the next step.
See section 2 for more details on the notion of angles.

⟨***options***⟩ see section 2.4.6 for the available ⟨*options*⟩. The most common option probably here is `/kata/dir/flip` → P. 9.

⟨***angleA***⟩ is where the movement starts (in degrees -360 to 360)

⟨***angleB***⟩ is where the movement ends (in degrees -360 to 360)

Note for ⟨*angleA*⟩ the string *center* is also eligible. This causes the arrow to start at the center of the circle/head, indicating only a direction not a turn.

`\kataKiai{⟨angle⟩}`

Use this command to draw an indicator that this step involves *kiai*.
The ⟨*angle*⟩ specifies at which angle to place this indicator. This package does not keep track how full the pictogram already is. Thus, you need to manually check at which angle there is space for this indicator to be placed.

2.3 Simple Example

Note: The output looks slightly different in the example's output here. Usually, the section is a proper section and the marginline is really show in the margin.

Mocks are only used here in the docs avoid problems showing this inside the example box <*doc>

A simple example without customizations (but includes flip)

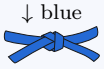
```
\begin{kata}[width=2]{Simple Example}{blue}
  Some explanation -- normal \LaTeX{} code
  \parindent=0pt

  \begin{step}
    \stance{stance A}{0}
    \waza{technique A}{-90}
    \dir{center}{90}
  \end{step}
```

```
  \begin{step}
    \stance{stance A}{0}
    \dir{90}{-90}
  \end{step}
  \begin{step}
    \stance{stance A}{0}[flip]
    \dir[flip]{90}{-90}
  \end{step}
\end{kata}
```

X. Simple Example

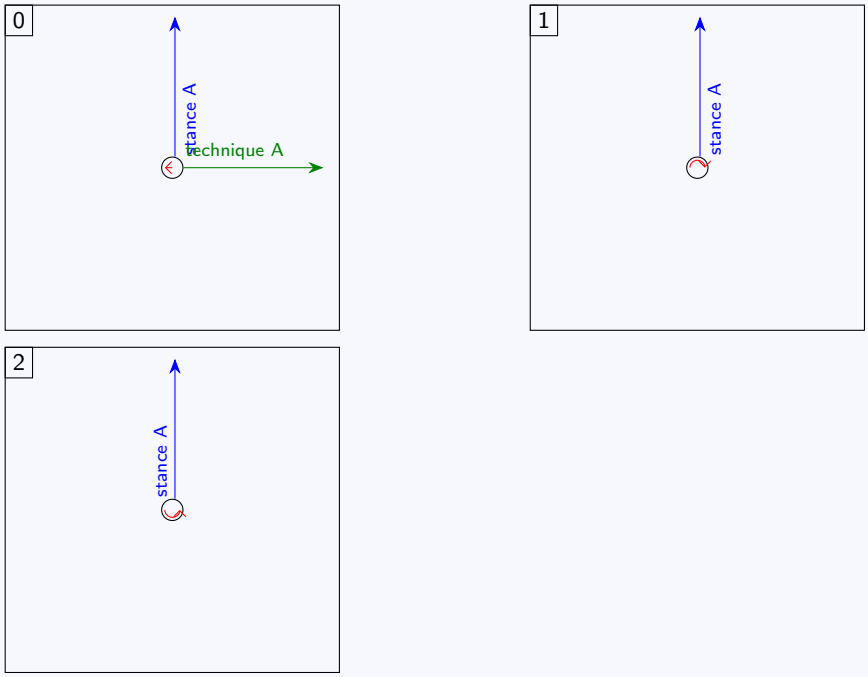
Margin would be:
(*begin marginline*)

↓ blue



(*end marginline*)

Some explanation – normal LaTeX code



</doc>

`\kataStepCnt{`⟨*counter*⟩`}`

Formats the counter holding the number of the current step. Adjust this with e.g. `\RenewDocumentCommand` or `renewcommand` to change how the counter is displayed.

2.4 Options / Customization

The idea is you can set any option with any command (offering generic ⟨*options*⟩) which makes sense at this place. For example you can set the style

in which the *border* of a step is drawn at the `kataStep`[→P. 3], `kata`[→P. 3] and `\kataSetup`[→P. 2]/package level all with varying scopes. But you cannot set the *border* style at the `\kataWaza`[→P. 3] or `\kataStance`[→P. 4] level as it does not make sense.

Note: You should never needs to specify paths manually when specifying options. As long as you are only setting options which are valid for that command, it suffices to specify the last component of the path (e.g. when setting `/kata/stance/flip`[→P. 9] in `\kataStance`[→P. 4], it suffices to speficy the option as `flip`).

Structuring the options, it became evident there are groups of commands/environments which should accept a common set of options following that logic. These are described in section 2.4.1. Sections 2.4.2 to 2.4.6 describe options specific to the commands/environments.

2.4.1 Common options

These options can be applied at `kataStep`[→P. 3], `kata`[→P. 3] and `\kataSetup`[→P. 2]/package level.

`/kata/setup/stance=`⟨*options*⟩
`/kata/kata/stance=`⟨*options*⟩
`/kata/step/stance=`⟨*options*⟩

Sets the ⟨*options*⟩ as if they are passed to every `\kataStance`[→P. 4] in the scope.
Scope means, e.g. if this is passed to `kata`[→P. 3], all `\kataStance`[→P. 4] commands in this kata will get these ⟨*options*⟩. But for the next kata, the ⟨*options*⟩ are not applied.

`/kata/setup/waza=`⟨*options*⟩
`/kata/kata/waza=`⟨*options*⟩
`/kata/step/waza=`⟨*options*⟩

Similar like `/kata/setup/stance` etc.
Sets the ⟨*options*⟩ as if they are passed to every `\kataWaza`[→P. 3] in the scope.

`/kata/setup/dir=`⟨*options*⟩
`/kata/kata/dir=`⟨*options*⟩
`/kata/step/dir=`⟨*options*⟩

Similar like `/kata/setup/stance` etc.
Sets the ⟨*options*⟩ as if they are passed to every `\kataDir`[→P. 4] in the scope.

`/kata/setup/color=`⟨*options*⟩
`/kata/kata/color=`⟨*options*⟩
`/kata/step/color=`⟨*options*⟩

Use a list of the options documented in section 2.4.1 as ⟨*options*⟩. Allows adjusting the colors of the various elements which are drawn.
Scoping works similar like described for `/kata/setup/stance` etc.

`/kata/setup/head style=`⟨*options*⟩
`/kata/kata/head style=`⟨*options*⟩
`/kata/step/head style=`⟨*options*⟩

Appends to the syle used for the circle representing the *head*. You can specify arbitrary Ti*k*Z-code here.
Scoping works similar like described for `/kata/setup/stance` etc.

`/kata/setup/border style=`⟨*options*⟩
`/kata/kata/border style=`⟨*options*⟩
`/kata/step/border style=`⟨*options*⟩

Similar like `/kata/setup/head style` → P. 6 etc.
Appends the ⟨*options*⟩ to the style for the border drawn around each pictogram.

`/kata/setup/border preset=`⟨*presets*⟩
`/kata/kata/border preset=`⟨*presets*⟩
`/kata/step/border preset=`⟨*presets*⟩

Similar like `/kata/setup/head style` → P. 6 etc. But intead of appending raw Ti*k*Z options, this option allows passing ⟨*presets*⟩ defined by this package for the border. See section 2.4.7 for more detailled information and a list of defined border presets.

`/kata/setup/step counter style=`⟨*options*⟩
`/kata/kata/step counter style=`⟨*options*⟩
`/kata/step/step counter style=`⟨*options*⟩

Similar like `/kata/setup/head style` → P. 6 etc.
Appends the ⟨*options*⟩ to the style for the Ti*k*Z node showing the number of the step (usually placed at the top left corner).

`/kata/setup/step phantom style=`⟨*options*⟩
`/kata/kata/step phantom style=`⟨*options*⟩
`/kata/step/step phantom style=`⟨*options*⟩

Similar like `/kata/setup/head style` → P. 6 etc.
Appends the ⟨*options*⟩ to the style activated when `/kata/step/phantom` → P. 8 is set to true (is also used automatically in `kata` → P. 3).

Colors A collection of keys used to adjust the color used for drawing elements in the pictograms. As ⟨*color*⟩, everything can be passed here which Ti*k*Z can use as a color.

`/kata/color/stance=`⟨*color*⟩

The color used for the arrow showing the direction of the stance as well as the label on this arrow.

`/kata/color/waza=`⟨*color*⟩

The color used for the arrow showing the direction of the waza as well as the label on this arrow.

`/kata/color/dir=`⟨*color*⟩

The color used for the arrow showing the next direction/turn for transitioning to the next step.

`/kata/color/head=`⟨*color*⟩

The color used for the circle representing the head.

`/kata/color/border`=⟨*color*⟩

The color used for the border around the step.

`/kata/color/step counter`=⟨*color*⟩

The color used for the node (border + content) showing the number of the step.

2.4.2 Package/Setup specific

These options are specific to the package-options and `\kataSetup`^{→P. 2}.

`/kata/setup/width`=⟨*width*⟩ (initially: 6)

Similar like `/kata/kata/width`.

`definealiases` (cmd-key)

Defines the same aliases like `\kataAliases`^{→P. 2} when set.
Defining the aliases is deferred until the begin of the document. Each alias is only defined if the corresponding macro is not already defined (otherwise a warning is shown in the log)
At the moment, this option only has an effect when given in the preamble (via `\kataSetup`^{→P. 2}) or in the package-options.

2.4.3 Kata environment

These options are specific to `kata`^{→P. 3}.

`/kata/kata/width`=⟨*width*⟩ (initially: 6)

Sets the amount of pictograms which fit one line in the document. As of now this package cannot measure this on its own so you need to set this manually (usually only once for the whole document via `/kata/setup/width`).
The effect of this is that at the end of a kata, a not fully populated line is filled up with `/kata/step/phantom` `kataStep`^{→P. 3}s. This is needed if you want a proper grid of pictograms in the output.

2.4.4 Step environment

These options are specific to `kataStep`^{→P. 3}.

`/kata/step//first step` (bool, initially: false, default: true)

Set this option on the first `kataStep`^{→P. 3} in a `kata`^{→P. 3}. As a result some spacing adjustments will be made.
Specifically this pushes the lines showing the `kataStep`^{→P. 3}s at the bottom of the page. The idea is that this makes reading easier when skipping from one page to another.
Beware, in the future additional adjustments might get coupled to this.

`/kata/step/phantom` (bool, initially: false, default: true)

Activates `/kata/step/step phantom style`^{→P. 7} for the most part.
The idea is that this makes the `kataStep`^{→P. 3} still occupy space while being invisble in the output. At the moment, this only works for empty steps (no manual content specified).

2.4.5 Typical content commands

Similarly like for global options there is a set of options which apply to all *typical* content commands. *Typical* here refers to content commands which draw something and also have a label. In particular `\kataDir` →P. 4 does not count as *typical* as it draw no label.

`/kata/waza/flip` (bool, initially: false, default: true)
`/kata/stance/flip` (bool, initially: false, default: true)

Flips the respective label to the other side.

`/kata/waza/style`
`/kata/stance/style`

Appends to the respective style of the component. Only adjusts the style for this single invocation (scoping).

`/kata/waza/label`
`/kata/stance/label`

Appends to the respective style for the label of the component. Only adjusts the style for this single invocation (scoping).
The style only applies to *non-flipped* labels.

`/kata/waza/label flipped`
`/kata/stance/label flipped`

Similar like `/kata/waza/label` etc.
The style only applies to *flipped* labels.

`/kata/waza/color`
`/kata/stance/color`

Sets the respective color for the component.
Under the hood this forwards to the respective key in `/kata/color/waza` →P. 7 etc.

Waza command no options apart from the common ones listed above.

Stance command no options apart from the common ones listed above.

2.4.6 Atypical content commands

Options for commands specifying content but which do not count as *typical* (see section 2.4.5).

Dir command

`/kata/dir/flip` (bool, initially: false, default: true)

Flips the arrow indicating a turn. Just specifying the angles for a turn is not enough to fully specify a turn. E.g. when turning from 0° to 180° the information whether to turn clockwise or counter-clockwise is still lacking. `\kataDir` →P. 4 will draw the arrow in one direction, it is the wrong one you can draw it in the other one with this option.
Has no effect if ⟨*angleA*⟩ is set to *center*.

Kiai command nothing additional

2.4.7 Presets

In order to make common but more complex style modifications easier, this package defines some *presets* which can usually be applied with the respective `<>` `preset` key.

Presets for the border

`/kata/preset/border/arrow up`
`/kata/preset/border/arrow down`
`/kata/preset/border/arrow right`
`/kata/preset/border/arrow left`

Adjust `/kata/step/border style` → P. 7 by appending so that the border shows two arrows. The arrows are placed on opposite sides of the rectangle both pointing in the same direction.
Which direction depends on which of the keys is chosen. The two sides showing the arrows are the following:

- edge on the *right* and *left* side for `/kata/preset/border/arrow up` and `/kata/preset/border/arrow down`
- edge on the *top* and *bottom* side for `/kata/preset/border/arrow right` and `/kata/preset/border/arrow left`

`/kata/preset/border/accent=`⟨*color*⟩

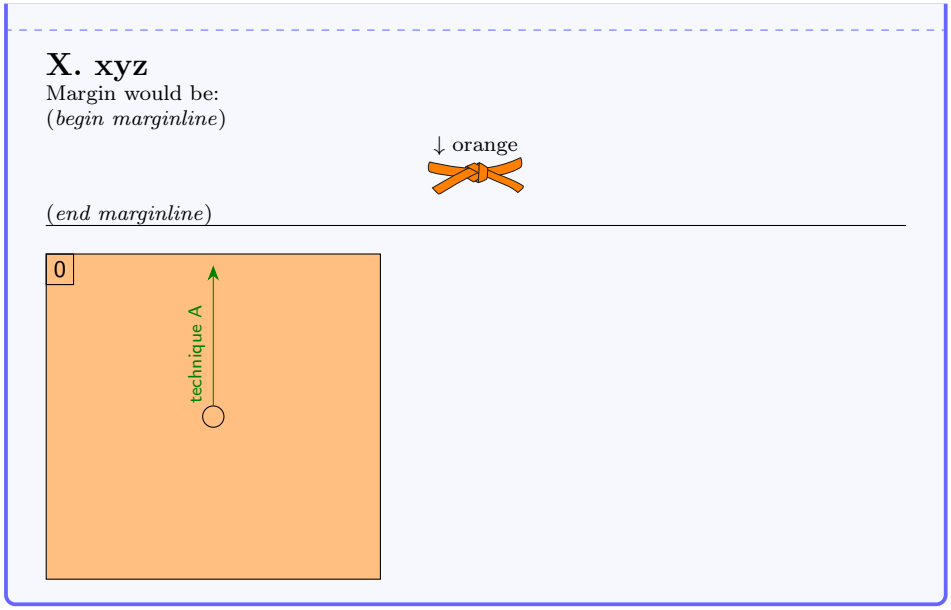
Colors the border with ⟨*color*⟩. Also fills the border with 5 % (transparent for the rest part) of that color to highlight/accent it.
As ⟨*color*⟩, everything can be passed here which Ti*k*Z can use as a color.

Custom presets Note: Styles (e.g. `/kata/waza/style` → P. 9) are evaluated in a `/tikz` context eventually. So if you want to declare a custom reusable style here is an example how you could declare a preset/style which only sets a background color (frame stays black) with less transparency:

Mocks are only used here in the docs avoid problems showing this inside the example box <*doc>

Idea how to manually define a reusable style

```
\tikzset{
  myPreset/.style={fill=#1!50},
}

\begin{kata}[width=2]{xyz}{orange}
  \parindent=0pt

  \begin{step}[border style={myPreset=orange}]
    \waza{technique A}{0}
  \end{step}
\end{kata}
```



</doc>

2.5 Extended Example

Note: The output looks slightly different in the example's output here. Usually, the section is a proper section and the marginline is really show in the margin.

Mocks are only used here in the docs avoid problems showing this inside the example box <*doc>

Extended example with all kinds of possible customizations

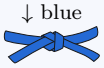
```
\begin{kata}[width=2,color={waza=orange}]{Extended Example}{blue}
  Some explanation -- normal \LaTeX{} code
  \parindent=0pt

  \begin{step}[color={stance=pink}]
    \stance{stance A}{0}
    \waza{technique A}{-90}[color=black]
    \dir{center}{90}
  \end{step}
  \kataSetup{color={dir=brown}}
  {
    \kataSetup{border preset={arrow up,accent=green}}
    \begin{step}[]
      \stance{stance B}{180}
      \waza{technique B}{90}
      \dir{90}{180}
    \end{step}
    \renewcommand\kataStepCnt[1]{\roman{#1}}
    \begin{step}
      \stance{stance B}{180}
      \waza[R]{technique B}{90}[flip]
      \dir[flip]{90}{180}
    \end{step}
  }
```

```
  \pgfkeys{
    /kata/style/step counter/.append style={xshift=1em},
    /kata/style/border/.append style={/kata/preset/helpers/arrowUD=45},
  }
  \begin{step}
    \stance{stance C}{180}
    \waza{technique D}{0}
  \end{step}
\end{kata}
```

X. Extended Example

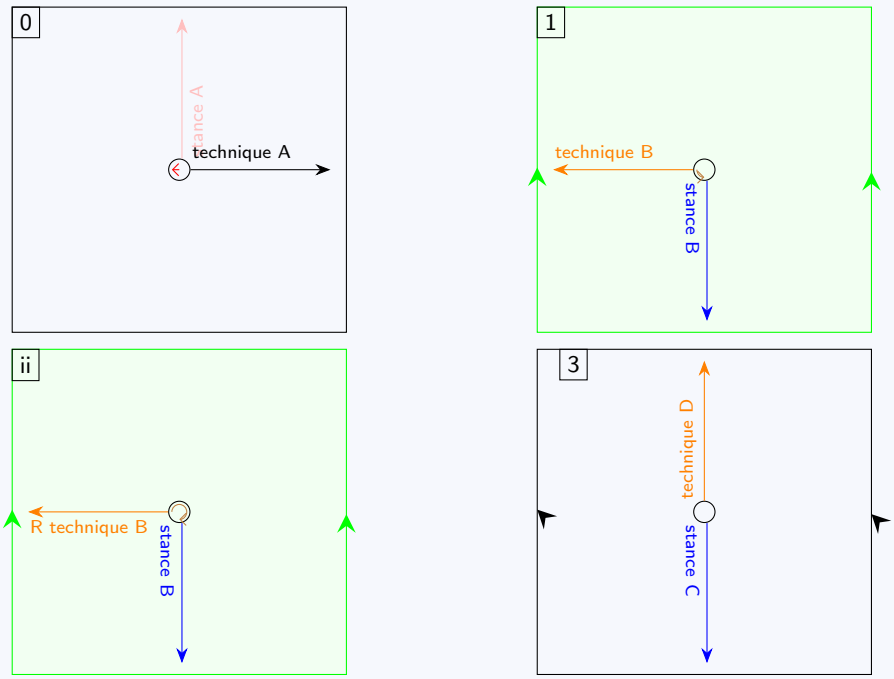
Margin would be:
(*begin marginline*)

↓ blue



(*end marginline*)

Some explanation – normal LaTeX code



</doc>

3 Implementation

```
\RequirePackage{memoize}
\RequirePackage{pgfkeys}
\RequirePackage{pgfopts}
\RequirePackage{coloredbelts}
\RequirePackage{tikz}
\usetikzlibrary{calc,positioning,fit,arrows.meta}
\usetikzlibrary{decorations.markings,decorations.pathmorphing,backgrounds}
```

Define marginline in case it isn't already defined (so the package works without KOMA-classes as well).

```
\AddToHook {begindocument} [kata/provideMarginline] {%
  \providecommand\marginline[1]{%
    \marginpar[{\raggedleft #1}]{\raggedright #1}%
  }%
}
```

Define own string compare for equality so the package works without KOMA as well.

```
\ExpandArgs {nnc} \UseName {cs_set_eq:NN} {\kata@ifstr} {str_if_eq:eeTF}
```

3.1 General pgfkeys setup

```
\pgfkeys{
```

for package wide settings

```
  /kata/setup/.is family,
```

for settings regarding for the kata env

```
  /kata/kata/.is family,
```

for settings regarding for the step env

```
  /kata/step/.is family,
```

for settings regarding for the stance command

```
  /kata/stance/.is family,
```

for settings regarding for the waza command

```
  /kata/waza/.is family,
```

for settings regarding for the dir command

```
  /kata/dir/.is family,
```

for settings regarding coloring

```
  /kata/color/.is family,
```

internal style definitions

```
  /kata/style/.is family,
```

presets for better usability

```
  /kata/preset/border/.is family,
```

forwards in order to DRY

```
  /kata/forward/.is family,
}
```

3.2 Set up pgfkeys-forwards

```
\pgfkeys{
  /kata/forward,
```

Common keys which simply delegate to other keys

```
  delegate common/.style={
    stance/.style = {/kata/stance,##1},
    waza/.style = {/kata/waza,##1},
    dir/.style = {/kata/dir,##1},
    color/.style = {/kata/color,##1},
  },
```

Common keys which append to styles

```
  append common/.style={
    head style/.style = {/kata/style/head/.append style = {##1}},
    border style/.style = {/kata/style/border/.append style = {##1}},
```

```
    border preset/.forward to = /kata/style/border preset,
    step counter style/.style = {/kata/style/step counter/.append style = {##1}},
    step phantom style/.style = {/kata/style/step phantom/.append style = {##1}},
  },
}
%
%
% \subsection{Package wide options}
% Defer keys in setup to the respective locations
%    \begin{macrocode}
\pgfkeys{
  /kata/setup,
  width/.style = {/kata/kata,width={#1}},
  definealiases/.code = {%
    \AddToHook {begindocument} [kata/aliases] {%
      \kataAliases%
    }%
  },
  /kata/forward/delegate common,
  /kata/forward/append common,
}
```

Setup package option parsing

```
\ProcessPgfPackageOptions{/kata/setup}
```

A setup command for setting package options later

\kataSetup

```
\NewDocumentCommand{\kataSetup}{m}{%
  \pgfkeys{/kata/setup,#1}%
  \ignorespaces%
}%
```

3.3 Optionally define command aliases

\kataAliases

```
\newif\ifkata@aliasesdefined
\kata@aliasesdefinedfalse
\NewDocumentCommand{\kataAliases}{}{
  \ifkata@aliasesdefined\else%
    \kata@aliasesdefinedtrue%
    \newcommand{\kata@warningdefined}[1]{\PackageWarning{kata}{Macro ##1 is already defined, di
    \@ifundefined{stance}{\let\stance\kataStance}{\kata@warningdefined{\string\stance}}%
    \@ifundefined{waza}{\let\waza\kataWaza}{\kata@warningdefined{\string\waza}}%
    \@ifundefined{dir}{\let\dir\kataDir}{\kata@warningdefined{\string\dir}}%
    \@ifundefined{kiai}{\let\kiai\kataKiai}{\kata@warningdefined{\string\kiai}}%
    % environment begin/end macros
    \@ifundefined{step}{%
      \let\step\kataStep%
      \let\endstep\endkataStep%
    }{%
      \PackageWarning{kata}{Environment/Macro \string\step is already defined, did not define t
    }%
  \fi%
}
```

3.4 Core internals

A counter for each pictogram to enable referencing.

```
\newcounter{kata@step}[section]
```

`\kata@requirestep` enforces something is run within the step environment (helper)

```
\newif\ifkata@instep
\newcommand*{\kata@requirestep}[1]{%
  \ifkata@instep\else
    \PackageError{kata}{#1 used outside step environment}{}
  \fi
}
```

`\kata@requirekata` enforces something is run within the kata environment (helper)

```
\newif\ifkata@inkata
\newcommand*{\kata@requirekata}[1]{%
  \ifkata@inkata\else
    \PackageError{kata}{#1 used outside kata environment}{}
  \fi
}

\newif\ifkata@stance@flip
\newif\ifkata@waza@flip
\newif\ifkata@dir@flip
\newif\ifkata@step@first
\newif\ifkata@step@phantom
\newcount\kata@tmpcnta
```

3.5 The kata environment

```
\pgfkeys{
  /kata/kata,
  width/.store in = \kata@width,
  width = 6,
  width/.value required,
  /kata/forward/delegate common,
  /kata/forward/append common,
}
```

`kata` (*env.*) this environment is basically a container for a sequence of steps. Arguments:

1. optional: keys from `/kata/kata`
2. kata name
3. for which belt / marginline
4. optional: belt color (for drawing), if empty -> not drawn, if not present use arg 3

```
\NewDocumentEnvironment{kata}{O{}mmo}{%
  \pgfkeys{/kata/kata,#1}%
  \section{#2}%
  \IfBlankTF{#3}{%
```

show if arg 3 is empty but arg 4 was set

```
      \IfNoValueF{#4}{\kata@beltmargin{}{#4}}%
    }{%
```

use arg 3 as fallback if arg 4 was not set

```
      \kata@beltmargin{#3}{\IfNoValueTF{#4}{#3}{#4}}%
    }%
    \setlength{\lineskip}{6pt}\noindent%
    \setcounter{kata@step}{0}%
    \kata@inkatatrue%
    \ignorespaces%
```

there is no harm in a single word/pictogram in the last line to turn off the corresponding penalty

```
    \widowpenalty=0
```

allow large larger spaces between the pictograms if this avoids drawing a picture outside of the textarea

```
    \sloppy
  }{%
```

add hidden steps to the last line. This ensures a uniform looking grid of pictograms

```
    \pgfmathtruncatemacro{\kata@missing}{mod(\kata@width - mod(\c@kata@step,\kata@width),\kata@wi
    \setcounter{kata@step}{0}%
    \ifnum\kata@missing>0
      \foreach \i in {1,...,\kata@missing} {%
        \begin{step}[phantom]\end{step} %
      }%
    \fi%
    \kata@inkatafalse%
```

avoid left-aligning the last line. Instead it should be justified in order to obtain a proper grid

```
    \parfillskip=0pt
    \par%
  }
```

`\kata@beltmargin`

```
  \newcommand{\kata@beltmargin}[2]{%
    \marginline{%
      \centering\footnotesize%
      \IfBlankF{#1}{%
        $\downarrow$ #1%
      }%
```

only linebreak if both are non-empty

```
    \IfBlankF{#1}{\IfBlankF{#2}{\\}}%
    \IfBlankF{#2}{%
        \ColorBelt*[scale=0.1]{#2}%
    }%
    }%
  }
```

3.6 The step environment

```
\pgfkeys{
  /kata/step,
  first step/.is if = kata@step@first,
  first step/.default = true,
  phantom/.is if = kata@step@phantom,
  phantom/.default = true,
  phantom = false,
  phantom/.append code={\ifkata@step@phantom\pgfkeysalso{/kata/style/step phantom}\fi},
  /kata/forward/delegate common,
  /kata/forward/append common,
}
```

`step` (*env.*) this environment always should wrap `kata@step`. It handles the aspects of positioning the pictogram. Arguments: 1 (optional): whether this is the first step of the kata / pgfkeys

```
\NewDocumentEnvironment{kataStep}{O{}+b}{%
  \kata@requirekata{\string\kataStep}%
  \pgfkeys{/kata/step,#1}%
  \ifkata@step@first\par\nointerlineskip\vfil\else\fi%
  \begin{kata@step}%
    #2
  \end{kata@step}%
  \ifkata@step@phantom\else\refstepcounter{kata@step}\fi%
}{}
```

`kata@step` (*env.*) the environment to draw everything else in

```
\newenvironment*{kata@step}{%
  \kata@requirekata{\string\step}%
  \begin{tikzpicture}[font=\scriptsize]%
    \begin{scope}[local bounding box=output]%
      \kata@head\kata@nothing%
      \kata@insteptrue%
}{
    \kata@instepfalse%
    \end{scope}%
```

Drawing the border on the background has the advantage the rectangle can be filled and colorize the background as well.

```
    \begin{scope}[on background layer]%
      \node[/kata/style/border] (t1) {};%
      \node[/kata/style/step counter] (cnt) {\kataStepCnt{kata@step}};%
    \end{scope}%
  \end{tikzpicture}%
}
```

3.7 Specify the Content

`\kiai` draw indicator for *kiai*. Arguments:

1. (optional) distance to head TODO: not implemented yet (probably will become pgf-interface)
2. angle

```
\NewDocumentCommand{\kataKiai}{O{}m}{
  \kata@requirestep{\string\kataKiai}
  \def\base{#2+90} \def\off{30} \def\ext{40} \def\roff{4pt} \def\len{0.75}
  \draw[shorten <=10pt,CtpRed]
  ($(head)+(\base-\off:\roff)$) edge[decorate,decoration={bent,aspect=0.3,amplitude=-2.5pt}] co
  ($(head)+(\base+\off:\roff)$) edge[decorate,decoration={bent,aspect=0.3,amplitude=+2.5pt}] co
  \path (tmpA) -- (tmpB) node[midway,sloped,font={\sffamily\scriptsize},CtpRed,draw,decorate,de
}
```

3.7.1 The stance command

```
\pgfkeys{
  /kata/stance,
  flip/.is if = kata@stance@flip,
  flip/.default = true,
  style/.style = {/kata/style/stance/.append style = {#1}},
  label/.style = {/kata/style/stance label/.append style = {#1}},
  label flipped/.style = {/kata/style/stance label flipped/.append style = {#1}},
  color/.forward to = /kata/color/stance,
}
```

`\kataStance` draw the arrow/path showing the stance Usage:

1. (optional) side to use (L/R) / prefix
2. name/text
3. angle
4. (optional) pgfkeys interface

```
\NewDocumentCommand{\kataStance}{o+mmO{}}{%
  \pgfkeys{/kata/stance,#4}%
  \kata@requirestep{\string\kataStance}%
  \pgfmathsetmacro{\kata@dir}{#3+90}%
  \pgfmathsetmacro{\x}{abs(sin(\kata@dir))}%
  \pgfmathsetmacro{\y}{-abs(cos(\kata@dir))}%
  \IfNoValueTF{#1}{\def\kata@nodecontent{}}{\def\kata@nodecontent{#1 }}%
  \draw[/kata/style/stance={\x}{\y}]
    (head) -- ++(\kata@dir:2)
    node[\ifkata@stance@flip/kata/style/stance label flipped\else/kata/style/stance label\fi]
    {\hangafter=1\settowidth\hangindent{\kata@nodecontent}\kata@nodecontent#2}
  ;
}
```

3.7.2 The waza command

```
\pgfkeys{
  /kata/waza,
  flip/.is if = kata@waza@flip,
  flip/.default = true,
  style/.style = {/kata/style/waza/.append style = {#1}},
  label/.style = {/kata/style/waza label/.append style = {#1}},
  label flipped/.style = {/kata/style/waza label flipped/.append style = {#1}},
  color/.forward to = /kata/color/waza,
}
```

`\kataWaza` draw the arrow/path showing a technique. Arguments:

1. (optional) side to use (L/R) / prefix
2. name/text
3. angle
4. (optional) pgfkeys interface

```
\NewDocumentCommand{\kataWaza}{o+mmO{}}{%
  \kata@requirestep{\string\kataWaza}%
  \pgfkeys{/kata/waza,#4}%
  \IfNoValueTF{#1}{\def\kata@nodecontent{}}{\def\kata@nodecontent{#1 }}%
  \pgfmathsetmacro{\kata@dir}{#3+90}%
  \draw[/kata/style/waza]
    (head) -- ++(\kata@dir:2)
    node[\ifkata@waza@flip/kata/style/waza label flipped\else/kata/style/waza label\fi]
    {\hangafter=1\settowidth\hangindent{\kata@nodecontent}\kata@nodecontent#2}
  ;
}
```

3.7.3 The dir command

```
\pgfkeys{
  /kata/dir,
  flip/.is if = kata@dir@flip,
  flip/.default = true,
  style/.style = {/kata/style/dir/.append style = {#1}},
  color/.forward to = /kata/dir/waza,
}
```

`\dir` draw a short arrow/path indicating the next turn/direction. Arguments:

1. (optional) flip / pgfkeys
2. source (Center,angle)
3. dst (angle)

```
\NewDocumentCommand{\kataDir}{O{}mm}{
  \kata@requirestep{\string\kataDir}
  \pgfkeys{/kata/dir,#1}
  \pgfmathsetmacro{\kata@out}{#3+90}
  \kata@ifstr{#2}{center}{
    \draw[/kata/style/dir] (head.center) -- ++(\kata@out:.33em);
  }{
    \pgfmathsetmacro{\kata@in}{#2+90}
    \pgfmathsetcount{\kata@tmpcnta}{\kata@out-\kata@in}
    \ifkata@dir@flip
      \ifnum\kata@tmpcnta<0
        \pgfmathsetmacro{\kata@outdeltaii}{360+\the\kata@tmpcnta}
      \else
        \pgfmathsetmacro{\kata@outdeltaii}{\the\kata@tmpcnta-360}
      \fi
    \else
      \pgfmathsetmacro{\kata@outdeltaii}{\the\kata@tmpcnta}
```

```
    \fi
    \draw[/kata/style/dir]
      (head.center) ++ (\kata@in:.33em)
      arc[start angle=\kata@in,delta angle=\kata@outdeltaii,radius=.33em]
    ;
  }
}
```

3.7.4 Internal elements

`\kata@nothing` ensure the box is always at least the same size (internal command + always placed)

```
\newcommand*{\kata@nothing}{
  \draw[draw=none] (head) -- ++(0:2.15) -- ++(180:4.3) -- ++(90:2.15) -- ++(-90:4.3);
}
```

`\kata@head` draw the circle symbolizing the head (internal command + always placed)

```
\newcommand*{\kata@head}{
  \node[/kata/style/head] (head) {};
}
```

3.8 Color keys

```
\pgfkeys{
  /kata/color,
  %
  stance/.store in = \kata@color@stance,
  stance/.value required,
  stance = blue,
  %
  waza/.store in = \kata@color@waza,
  waza/.value required,
  waza = green!50!black,
  %
  dir/.store in = \kata@color@dir,
  dir/.value required,
  dir = red,
  %
  head/.store in = \kata@color@head,
  head/.value required,
  head = black,
  %
  border/.store in = \kata@color@border,
  border/.value required,
  border = black,
  %
  step counter/.store in = \kata@color@stepcounter,
  step counter/.value required,
  step counter = black,
}
```

3.9 Styles

3.9.1 Styles for drawn elements

```
\pgfkeys{
```

```
  /kata/style,
  %
  head/.style = {
    circle,
    draw=\kata@color@head,
    inner xsep=0.333em,
  },
  %
  stance/.style 2 args = {
    draw=\kata@color@stance,
    -{Stealth[scale=1.2]},
    transform canvas={xshift=#1,yshift=#2},
  },
  %
  waza/.style = {
    draw=\kata@color@waza,
    -{Stealth[scale=1.2]},
  },
  %
  dir/.style = {
    draw=\kata@color@dir,
    -{Straight Barb},
  },
  %
  border/.style = {
    rectangle,
    draw=\kata@color@border,
    fit=(output),
    inner sep=0pt,
    outer sep=0pt,
    inner xsep=0.2em,
  },
  %
  border preset/.code={
    \pgfkeysalso{/kata/preset/border,#1}%
  },
  %
  step counter/.style = {
    draw=\kata@color@stepcounter,
    font=\sffamily\small,
    outer sep=0pt,
    at={(t1.north west)},
    below right,
    rectangle,
  },
}
```

3.9.2 Styles for labels

```
\pgfkeys{
  /kata/style,
  %
  stance label base/.style = {
    \kata@color@stance,
    midway,
```

```
    sloped,
    font=\sffamily\scriptsize,
    text width=1.8cm,
  },
  stance label/.style = {
    /kata/style/stance label base,
    below,
  },
  %
  stance label flipped/.style = {
    /kata/style/stance label base,
    above,
  },
  %
  waza label base/.style = {
    \kata@color@waza,
    midway,
    sloped,
    font=\sffamily\scriptsize,
    text width=1.8cm,
  },
  %
  waza label/.style = {
    /kata/style/waza label base,
    above,
  },
  %
  waza label flipped/.style = {
    /kata/style/waza label base,
    below,
  },
}
```

3.9.3 Styles for steps

```
\pgfkeys{
  /kata/style,
  %
  step phantom/.style = {
    /kata/style/border/.style={draw=none},
    /kata/style/head/.style={draw=none},
    /kata/style/step counter/.style={opacity=0},
  },
}
```

3.9.4 Presets for styles

Helpers

```
\pgfkeys{
  /kata/preset/helpers/.cd,
  arrowRL/.style={
    postaction={decorate},decoration={
      markings,
      mark=at position 0.125
        with {\pgftransformrotate{#1 - 270}\arrow[line width=2pt]{stealth}},
      mark=at position 0.625
```

```
        with {\pgftransformrotate{#1 - 270}\arrowreversed[line width=2pt]{stealth}},
    },
  },
  arrowUD/.style={
    postaction={decorate},decoration={
      markings,
      mark=at position 0.375
        with {\pgftransformrotate{#1}\arrow[line width=2pt]{stealth}},
      mark=at position 0.875
        with {\pgftransformrotate{#1}\arrowreversed[line width=2pt]{stealth}},
    },
  },
}
```

Border presets

```
\pgfkeys{
  /kata/preset/border,
  arrow right/.style={
    /kata/style/border/.append style={
      /kata/preset/helpers/arrowRL=-90,
    },
  },
  arrow left/.style={
    /kata/style/border/.append style={
      /kata/preset/helpers/arrowRL=+90,
    },
  },
  arrow up/.style={
    /kata/style/border/.append style={
      /kata/preset/helpers/arrowUD=0,
    },
  },
  arrow down/.style={
    /kata/style/border/.append style={
      /kata/preset/helpers/arrowUD=180,
    },
  },
  accent/.style={
    /kata/style/border/.append style={
      fill=#1!5,
      draw=#1,
    },
  },
}
```

3.10 Misc

`\kataStepCnt` Formats the counter for the step indicator

```
\NewDocumentCommand{\kataStepCnt}{m}{\arabic{#1}}
```

3.11 Memoization

TODO: which pgfkeys to add TOOD: make optional depending on whether memoize is loaded or not

```
\mmzset{auto={kata@step}{
  memoize,
  csname meaning to context={environment kata@step\space code},
  csname meaning to context={environment kata@step end aux\space},
  csname meaning to context={nothing code},
  csname meaning to context={head code},
  csname meaning to context={stance code},
  csname meaning to context={waza code},
  csname meaning to context={dir code},
}}
```

4 Change History

0.1.0

General: Converted to DTX file . . 1

5 Index

Numbers written in italic refer to the page where the corresponding entry is described; numbers underlined refer to the definition; numbers in roman refer to the pages where the entry is used.

Symbols

\@ifundefined 14
\\ 16

A

accent key 10
\AddToHook 13, 14
\arabic 23
\arrow 22, 23
arrow down key 10
arrow left key 10
arrow right key 10
arrow up key 10
\arrowreversed 23

B

\base 18
\begin 14, 16, 17
border key 8

C

\c@kata@step 16
\centering 16
\ColorBelt 16

D

\def 18, 19
definealiases key 8
dir key 7
\dir 14, 19
\downarrow 16
\draw 18–20

E

\else 14, 15, 17–19
\end 16, 17
\endkataStep 14
\endstep 14
environments:
  kata 15
  kata@step 17
  step 17
\ExpandArgs 13
\ext 18

F

\fi 14–20
first step key 8
flip key 9
\footnotesize 16

\foreach 16

H

\hangafter 18, 19
\hangindent 18, 19
head key 7

I

\i 16
\IfBlankF 16
\IfBlankTF 15
\ifkata@aliasesdefined 14
\ifkata@dir@flip 15, 19
\ifkata@inkata 15
\ifkata@instep 15
\ifkata@stance@flip 15, 18
\ifkata@step@first 15, 17
\ifkata@step@phantom 15, 17
\ifkata@waza@flip 15, 19
\IfNoValueF 16
\IfNoValueTF 16, 18, 19
\ifnum 16, 19
\ignorespaces 14, 16

K

kata (env.) 15
kata/border preset key 7
kata/border style key 7
kata/color key 6
kata/dir key 6
kata/head style key 6
kata/stance key 6
kata/step counter style key 7
kata/step phantom style key 7
kata/waza key 6
\kata@aliasesdefinedfalse 14
\kata@aliasesdefinedtrue 14
\kata@beltmargin 16, 16
\kata@color@border 20, 21
\kata@color@dir 20, 21
\kata@color@head 20, 21
\kata@color@stance 20, 21
\kata@color@stepcounter 20, 21
\kata@color@waza 20–22
\kata@dir 18, 19
\kata@head 17, 20
\kata@ifstr 13, 19
\kata@in 19, 20
\kata@inkatafalse 16
\kata@inkatatrue 16
\kata@instepfalse 17
\kata@insteptrue 17
\kata@missing 16
\kata@nodecontent 18, 19
\kata@nothing 17, 20
\kata@out 19
\kata@outdeltaii 19, 20
\kata@requirekata 15, 17
\kata@requirestep 15, 18, 19
kata@step (env.) 17
\kata@tmpcnta 15, 19
\kata@warningdefined 14
\kata@width 15, 16
\kataAliases 14, 14
\kataDir 14, 19
\kataKiai 14, 18
\kataSetup 14
\kataStance 14, 18
\kataStep 14, 17
\kataStepCnt 17, 23
\kataWaza 14, 19
Keys
definealiases 8
/kata/
kata/border preset 7
kata/border style 7
kata/color 6
kata/dir 6
kata/head style 6
kata/stance 6
kata/step counter style 7
kata/step phantom style 7
kata/waza 6
setup/border preset 7
setup/border style 7
setup/color 6
setup/dir 6
setup/head style 6
setup/stance 6
setup/step counter style . . . 7
setup/step phantom style . . . 7
setup/waza 6
stance/color 9
stance/flip 9
stance/label 9
stance/label flipped 9
stance/style 9
step/border preset 7
step/border style 7
step/color 6
step/dir 6
step/head style 6
step/stance 6
step/step counter style 7
step/step phantom style 7
step/waza 6
waza/color 9
waza/flip 9

waza/label 9
waza/label flipped 9
waza/style 9
/kata/color/
border 8
dir 7
head 7
stance 7
step counter 8
waza 7
/kata/dir/
flip 9
/kata/kata/
width 8
/kata/preset/border/
accent 10
arrow down 10
arrow left 10
arrow right 10
arrow up 10
/kata/setup/
width 8
/kata/step/
phantom 8
/kata/step//
first step 8
\kiai 14, 17

L

\len 18
\let 14
\lineskip 16

M

\marginline 13, 16
\marginpar 13
\mmzset 24

N

\newcommand 14–16, 20
\newcount 15
\newcounter 15
\NewDocumentCommand . . . 14, 18, 19, 23
\NewDocumentEnvironment 15, 17
\newenvironment 17
\newif 14, 15
\node 17, 20
\noindent 16
\nointerlineskip 17

O

\off 18

P

\PackageError 15
\PackageWarning 14
\par 16, 17
\parfillskip 16
\path 18
\pgfkeys 13–15, 17–23
\pgfkeysalso 17, 21
\pgfmathsetcount 19
\pgfmathsetmacro 18, 19
\pgfmathtruncatemacro 16
\pgftransformrotate 22, 23
phantom key 8
\ProcessPgfPackageOptions 14
\providecommand 13

R

\raggedleft 13
\raggedright 13
\refstepcounter 17
\RequirePackage 12
\roff 18

S

\scriptsize 17, 18, 22
\section 15
\setcounter 16
\setlength 16
\settowidth 18, 19
setup/border preset key 7
setup/border style key 7
setup/color key 6
setup/dir key 6
setup/head style key 6
setup/stance key 6
setup/step counter style key 7
setup/step phantom style key 7
setup/waza key 6
\sffamily 18, 21, 22
\sloppy 16
\small 21
\space 24
stance key 7
\stance 14
stance/color key 9
stance/flip key 9
stance/label key 9
stance/label flipped key 9
stance/style key 9
\step 14, 17
step (env.) 17
step counter key 8
step/border preset key 7
step/border style key 7
step/color key 6
step/dir key 6

step/head style key 6
step/stance key 6
step/step counter style key 7
step/step phantom style key 7
step/waza key 6
\string 14, 17–19
\subsection 14

T

\the 19

U

\UseName 13
\usetikzlibrary 12

V

\vfil 17

W

waza key 7
\waza 14
waza/color key 9
waza/flip key 9
waza/label key 9
waza/label flipped key 9
waza/style key 9
\widowpenalty 16
width key 8

X

\x 18

Y

\y 18